# Branch-Aware Scalar Certification for Cubic-Torus Isoperimetry

Local Moments, Signed Remainders, and Between-Knot Bounds

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**Abstract**

The pinned OpenAI manuscript on cubic-flat-three-torus isoperimetry reduces its last candidate exclusion to positivity of two explicitly defined scalar functions. We reconstruct that numerical dependency, including the three layer-cake branches, the capped nesting gain, and the distributional curvature bounds needed between knots. Local Ram moment expansion yields a signed fourth-moment enclosure for the sole nonlinear branch, without moving its endpoints; panel subdivision has an explicit fourth-order bound. A sharper gain curvature constant and endpoint-sensitive quadratic certificates strengthen the existing interpolation argument. An independently authored integer-interval run reproduces all 24 baseline knots, 16 gain knots, and 23 source interval bounds. Using the same rounded knots, the revised uniform margin exceeds 0.000280 instead of 0.000090; four-panel moment enclosures raise it above 0.000311. These are local analytic and arithmetic refinements, not a new proof of the geometric reduction or a historical novelty claim. The complete geometric classification remains conditional on the imported geometric premises. Contribution assessment: C2/5, proved branch-aware remainder and interpolation refinements with executed scalar certificates give modest local robustness improvements, while the global geometric reduction remains imported.

# Contents


**1 Scope, provenance, and claim types** **3**

**2 The geometric dependency, with normalizations** **3**
2.1 The source claim and the two endpoints . . . 3
2.2 Candidate reduction and section identities . . . 4

**3 The source-specific scalar functions** **5**
3.1 Why nesting, not just the planar profile, matters . . . 5
3.2 Layer cake and exact branch locations . . . 6

**4 Local moments and signed remainders** **7**
4.1 The actual private-corpus connection . . . 7
4.2 A branch-panel theorem . . . 7
4.3 Exact moments and cancellation control . . . 8

**5 Curvature bounds that survive switches** **8**
5.1 The baseline has a logarithmic concavity correction . . . 8
5.2 A smaller constant for the gain . . . 9

**6 Continuous certificates from finite knot enclosures** **10**
6.1 A logarithmic lower envelope . . . 10
6.2 A rational endpoint-sensitive alternative . . . 10
6.3 Bernstein form without pretending the integrand is polynomial . . . 10
6.4 Error ledger and conditional geometric conclusion . . . 11

**7 Independent arithmetic: protocol and results** **11**
7.1 What is and is not being executed . . . 11
7.2 Arithmetic design . . . 11
7.3 All interval margins, not selected samples . . . 12
7.4 Independent symbolic and diagnostic checks . . . 13
7.5 Independent geometric review and platform record . . . 14

**8 Benefits, limitations, and research directions** **14**

**9 Public literature check and assessment** **15**

**A Read locators and reproducibility contract** **15**

# 1 Scope, provenance, and claim types

This is a confidential, locally prepared technical report. Public checks use only public problem titles and URLs. Private passages are read locally and are not sent as search queries. The object of comparison is OpenAI's September 24, 2026 manuscript [1], at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The source's main theorem asserts the complete cubic-torus isoperimetric profile and all equality cases. That assertion is not automatically a theorem of this report. A separate auditor is responsible for the full geometric reduction. All substantive source sections and both appendices have been read, not merely their headings or abstract.

We distinguish five statuses. *Imported* statements are source results used with their stated hypotheses. *Proved here* statements have full local proofs below; this label means proved in this report, not historically new. *Conditional* consequences explicitly retain geometric or computational premises. *Executed* statements refer only to recorded runs and their checks. *Research directions* do not contribute to the score. The screening rating L5 records strong applicability of rigorous scalar integration, and is independent of the contribution rating C2.

The main realized analytic changes are a signed, weight-aware error in Section 4, a branch-safe improved curvature bound in Section 5, and an endpoint-sensitive certificate in Section 6. The source already uses Taylor expansion, moment recurrences, outward arithmetic, and semiconcavity. Merely renaming those steps as Ram calculus would contribute nothing. Our comparison is against that existing machinery.

# 2 The geometric dependency, with normalizations

## 2.1 The source claim and the two endpoints

On the unit torus, with $v = \min(V, 1-V)$, the claimed profile is

$$I_{\mathrm{unit}}(V) = \min\{(36\pi)^{1/3} v^{2/3},\ 2\sqrt{\pi v},\ 2\}. \tag{2.1}$$

The source works instead on $T = \mathbb{R}^3/(2\mathbb{Z}^3)$, with $V = |E|/8$, $A = |\partial E|/8$, and sum curvature $h = k_1 + k_2$. Thus $A = I_{\mathrm{unit}}/2$; $h$ is not divided by eight. In the reflected cube $[0,1]^3$, normalized area and volume become relative perimeter and ordinary volume. A period-two ball has $V = \pi R^3/6$, $A = \pi R^2/2$; a primitive axial tube of core length $2k$ has $V = k\pi R^2/4$, $A = k\pi R/2$. Since $k \geq 1$, the standard competitor envelope is

$$F(V) = \min\{(9\pi/2)^{1/3} v^{2/3}, \sqrt{\pi v}, 1\}. \tag{2.2}$$

Equating its neighboring branches gives $V_1 = 4\pi/81$ and $V_2 = 1/\pi$. At their upper branch endpoints, the ball and tube radii are $2/3$ and $2/\pi$, below the period-two injectivity radius one. Slabs have two coordinate tori and normalized area one. The asserted equality types are balls, shortest-geodesic tubes, coordinate slabs, and complements, with both adjacent types at the transitions.

The imported flat-manifold structure makes a nonstandard minimizing boundary connected of genus $g \geq 2$, with $h^2 A \leq \pi$ (genus four has $h = 0$). The parallel-surface upper support $C$ obeys

$$C' = h, \qquad C^2 C'' = -h^2 C + \pi(1-g). \tag{2.3}$$

For example, $w' = A$, $a' = hA$, and $a'' = \frac{1}{8}\int 2K$ for parallel displacement; applying the chain rule to $C(w) = a$ and Gauss–Bonnet gives (2.3). On each smooth branch of $F$, $F'' + (F')^2/F \geq 0$. If

$I - F$ has a negative interior minimum between consecutive transition points, its upper support has $C = I < F$, $C' = F'$, $C'' \geq F''$, contrary to (2.3). The same comparison at equality excludes nonstandard minimizers inside smooth branches. Endpoint co-minimizers must still be excluded separately; this distinction is material.

### 2.2 Candidate reduction and section identities

The source's reflection argument uses volume bisection, perimeter averaging, analytic continuation, and agreement of occupied sides. Stability applied to the odd Jacobi fields $N_i$ then gives strict coordinate monotonicity after possible coordinate reversal. The two surviving occupied vertex patterns are

$$\{000, 100, 010\}, \qquad \{000, 100, 010, 001\}.$$

This imports a global statement about all minimizers, not a finite sampling assumption. It also requires the source's analytic continuation and constrained nodal-domain arguments, which scalar computation does not test.

Write $v(z) = |E_z|$, $s = v(0)$, $r = v(1)$, $p(z)$ for the relative slice perimeter, $\alpha = \sqrt{1 - N_z^2}$, and $b_z = A^{-1} \int_{\Sigma \cap Q} (1 - N_z^2)$. The source derives

$$\int_{\partial E_z} \alpha \, d\ell = b_z A + h(v(z) - V), \tag{2.4}$$

$$s - r \leq A\sqrt{1 - b_z}, \qquad b_x + b_y + b_z = 2, \tag{2.5}$$

$$-v'(z) = \int_{\partial E_z} \frac{N_z}{\alpha} \, d\ell, \qquad A = \int_0^1 \int_{\partial E_z} \frac{1}{\alpha} \, d\ell \, dz. \tag{2.6}$$

The first equation follows by testing first variation with $\eta(z)e_z$ and averaging the resulting constant in $z$. The second is projection plus Cauchy–Schwarz. The section-area function is absolutely continuous: all critical levels lie on finitely many transverse axial-edge intersections, and continuity excludes atoms of its monotone derivative. Thus the area-coordinate substitution

$$\int_0^1 \gamma(v(z))(-v'(z)) \, dz = \int_r^s \gamma(w) \, dw$$

is valid for every nonnegative measurable $\gamma$, even when slice densities become unbounded at exceptional heights.

For the four-vertex pattern choose $b_z \geq 2/3$, set $d = \sqrt{1 - b_z}$ and $x = hA$. The source reduces its exclusion to

$$0 < d \leq 1/\sqrt{3}, \quad 0 \leq x \leq \sqrt{\pi}, \quad \tau + \frac{x}{\pi}(1 - \tau^2) = 1 - d^2,$$

and proves $d + [\tau^2 + (\tau + xd)^2]/\pi < 1$, contradicting a necessary bound in the opposite direction. Its elementary appendix first forces $\tau > 1/10$, then treats $\tau \leq 3/10$ by increasing $d$ to $1/\sqrt{3}$ and bounding a decreasing endpoint expression. For $\tau \geq 3/10$, setting $U = \sqrt{1 - \tau}$, $Z = d/U$ separates a left side greater than $9/8$ from a right side less than $9/8$. No numerical verifier is required for those displayed rational inequalities, and they are not a new contribution here.

For the three-vertex pattern the top collar is vacant. Equation (2.4) gives $b_z A = hV$, hence

$$\int \alpha \, d\ell = hv, \quad hs \geq \sqrt{\pi(1 - s)}, \quad s^2 \leq A(A - hV). \tag{2.7}$$

The first inequality forces $h > 0$; concavity and complementary symmetry of the profile then place this pattern on the smaller-volume side. At $V_1$, with $a = 2\pi/9$, the source obtains the impossible condition

$$1 \le a\sqrt{1 - x/\pi} + (x^2/\pi)(1 - x/\pi), \quad 0 \le x \le \pi\sqrt{2}/3.$$

Splitting at $x = 1$ gives upper bounds $58/63$ and $902/945$. At $V_2$, $A \le 1$ and (2.7) instead imply

$$\frac{\sqrt{\pi(1-s)}}{s} \le h \le \pi(1 - s^2), \quad .535 < s < .89, \quad .64 \le h \le 2.25, \quad s \ge t(h), \tag{2.8}$$

where

$$t(h) = \frac{2}{1 + \sqrt{1 + 4h^2/\pi}}, \qquad h^2t^2 = \pi(1 - t). \tag{2.9}$$

The $s$ bounds follow from the unimodality of $s^2(1 - s)(1 + s)^2$ and rational endpoint comparisons. This is the exact geometric entry point for our scalar work.

# 3 The source-specific scalar functions

All decimals below denote exact rationals. Set

$$\lambda = .98, \quad \ell = .28, \quad D = \sqrt{48}/7, \quad a_0 = 1/\pi, \quad b_0 = 1 - 1/\pi. \tag{3.1}$$

For $X \ge 0$, define $G(X) = g(\min(X, D))$ with

$$g(X) = \frac{1 - \lambda X^2}{\sqrt{1 - X^2}}, \qquad W(X) = -G'(X) = \frac{X(.96 - .98X^2)}{(1 - X^2)^{3/2}} \quad (0 \le X \le D).$$

Then $G(D) = \ell$, $W \ge 0$, and $\int_0^D W = 1 - \ell = .72$. The planar profile lower bound is

$$p_*(v) = \min\{\sqrt{\pi v}, 1, \sqrt{\pi(1 - v)}\}.$$

The baseline and nesting gain are

$$B(h) = \lambda h/\pi + \int_0^{t(h)} G(hv/p_*(v))\, dv - 1, \tag{3.2}$$

$$S(h) = \inf_{y \ge 0}\{\ell y^2 + C_0(h)\min(d_m, (P - Qy)_+)^3\}, \tag{3.3}$$

$$u = \min(b_0, .96/h), \quad P = u - 1 + 2ht/\pi, \quad d_m = .21,$$

$$Q = \frac{2}{\pi}\sqrt{\pi + 2h^2t}, \qquad C_0 = \min(.134h^3, .36).$$

$B$ is used on $[.64, 2.25]$, whereas $S$ is used only on $[1.02, 1.62]$. Extending the gain outside its proved geometric domain is not justified by the scalar formula alone.

## 3.1 Why nesting, not just the planar profile, matters

For $J \in [0, 1]$, $f_J(X) = X^{-1} - \lambda X - J\sqrt{1 - X^2}/X$ is convex, because

$$X^3 f_J''(X) = 2 - J\frac{2 - 3X^2}{(1 - X^2)^{3/2}} \ge 0.$$

Jensen with mean $hv/p$ and $J = G(hv/p)$ gives the source's integrated flux inequality; after coarea it yields $A - 1 \geq B(h) + \ell(s - t)$. The baseline becomes negative in the central range, so no refinement of quadrature can replace the missing geometric nesting gain.

Let $L_x, L_y$ be the extents of the bottom complement. Pairing its monotone tangent with the unit vector $(\eta/L, -\sqrt{1 - \eta^2/L^2})$ proves, for either extent,

$$hs \geq (1 - s)/L + \pi L/4.$$

Consequently a contained strip has smaller endpoint at most

$$1 - \frac{2}{\pi}\{hs - \sqrt{h^2 s^2 - \pi(1 - s)}\} \leq 1 - 2ht/\pi + Q\sqrt{s - t}.$$

Writing $Y = \sqrt{s - t}$ and $\delta = \min(d_m, (P - QY)_+)$ puts the relevant area interval $[u - \delta, u]$ inside $[a_0, b_0] \cap [0, t]$, with $hu \leq .96 < D$. On that interval, all three slice types obey

$$p^2 \geq 1 + c\delta^2 H(q), \quad c = 2.7, \quad H(q) = \min\{q^2, \pi(1 - q)/(cd_m)\}, \quad q = (v - u + \delta)/\delta.$$

For strips the source calibrates their graph with $(\sqrt{1 - k^2x^2}, kx)$, $k = 2.5\delta q$, obtaining $p \geq 1 + 1.35\delta^2 q^2$. Occupied corners use $p^2 \geq \pi v$; complementary corners use $p^2 \geq \pi(1 - v)$. This is why both strip orientations and the vanishing of $H(1)$ are essential.

The branches of $H$ meet at $\theta = 2/(1 + \sqrt{1 + 4cd_m/\pi})$. Their exact moments are

$$M = \frac{\theta^2(3 - \theta)}{6}, \quad q_m = \frac{2 + 2\theta - \theta^2}{2(3 - \theta)}, \quad p_2 = 1 + cd_m^2\theta^2.$$

The source proves $M > .2659$, $q_m > .697$, $p_2 < 1.09$. Its derivative lower bound $.91X$ on $0 \leq X \leq .96$ gives the additional area

$$\frac{.91c\delta^3}{p_2^{3/2}} \int_0^1 [h(u - \delta + \delta q)]^3 H(q)\, dq \geq C_0\delta^3.$$

Thus, on the interior curvature range, $A - 1 \geq B(h) + S(h)$. All the preceding geometrical inequalities are imported with explicit derivations to identify dependencies, not credited as our results.

### 3.2 Layer cake and exact branch locations

Let $\rho$ invert $r \mapsto r/\sqrt{\pi(1 - r)}$ on $[0, 1)$:

$$\rho(k) = \sqrt{\pi}\, k\sqrt{1 + \sigma k^2} - \frac{\pi}{2}k^2, \qquad \sigma = \pi/4, \quad k \geq 0. \tag{3.4}$$

Then $t = \rho(1/h)$ and

$$B(h) + 1 = \lambda h/\pi + \ell t + \int_0^D W(X) \min\{\pi(X/h)^2, X/h, \rho(X/h)\}\, dX. \tag{3.5}$$

Indeed $hv/p_*(v) = \max\{h\sqrt{v/\pi}, hv, hv/\sqrt{\pi(1 - v)}\}$. Its sublevel set has length equal to the displayed minimum. The restriction $v \leq t$ is redundant for $X \leq D < 1$. Tonelli applied to $G(q) = \ell + \int W(X)\mathbf{1}_{q \leq X}\, dX$ proves (3.5).

The selected $k = X/h$ branches are quadratic for $k \leq a_0$, linear for $a_0 \leq k \leq b_0$, and $\rho$ for $k \geq b_0$. Set $x = ha_0$ and $z = \min(D, hb_0)$; then

$$B + 1 = \lambda h/\pi + \ell t + \frac{\pi}{h^2}\int_0^x X^2 W\, dX + \frac{1}{h}\int_x^z XW\, dX + \int_z^D W(X)\rho(X/h)\, dX. \tag{3.6}$$

Only the last integral needs approximation. No polynomial approximation is used across a branch switch, and a zero-length last interval is omitted exactly.

# 4 Local moments and signed remainders

## 4.1 The actual private-corpus connection

The local differential representation in [5], at `thm:ram-local`, states a finite Taylor expansion under integrable absolute moments and $C^{N+1}$ regularity on every kernel segment, with remainder bounded by $B_{N+1}M_{N+1}/(N+1)!$. Here the nonnegative kernel is the fixed-$h$ measure $W(X)\,dX$ restricted to a branch panel; the signal is $\rho(X/h)$. Its support is compact and lies below $D<1$, so the moments exist. This is a pointwise forward calculation, not an inversion of a variable-coefficient differential operator.

The same private chapter explicitly distinguishes polynomial exactness from polynomial closure. Its degree-closure hypothesis does not apply to the $h$-dependent radicals here. In particular, no nilpotent finite inverse and no global polynomial representation of $B$ or $S$ follows. The private Bernstein chapter [6] applies to the polynomial enclosure after its residual has been retained, not directly to these nonpolynomial functions. The residual-aware gap principle in [7] supplies an error-accounting discipline; no SOS Gram certificate is asserted or needed.

## 4.2 A branch-panel theorem

For fixed $h\in[.64,2.25]$, choose a nonempty panel $[A,B]\subset[hb_0,D]$. Put $m=(A+B)/(2h)$ and

$$\mu_j=\int_A^B W(X)(X/h-m)^j\,dX,\qquad T_3(k;m)=\sum_{j=0}^{3}\frac{\rho^{(j)}(m)}{j!}(k-m)^j.$$

Let $I=\int_A^B W(X)\rho(X/h)\,dX$ and $I_3=\sum_{j=0}^3\rho^{(j)}(m)\mu_j/j!$.

**Theorem 4.1** (Signed, weight-aware enclosure)**.** *For every panel as above, let*

$$M_-=|\rho''''(B/h)|,\qquad M_+=|\rho''''(A/h)|.$$

*Then* $0<M_-\le M_+<4$ *and*

$$I_3-\frac{M_+\mu_4}{24}\le I\le I_3-\frac{M_-\mu_4}{24}\le I_3. \tag{4.1}$$

*In particular* $I_3-\mu_4/6\le I\le I_3$. *These inequalities include* $A=hb_0$ *and* $B=D$.

*Proof.* Differentiation of (3.4) gives

$$\rho'''(k)=\frac{3\sqrt{\pi}\sigma}{(1+\sigma k^2)^{5/2}},\qquad \rho''''(k)=-\frac{15\sqrt{\pi}\sigma^2k}{(1+\sigma k^2)^{7/2}}.$$

The derivative of the positive magnitude has the sign of $1-6\sigma k^2$, which is negative for $k\ge b_0$. The source's $|\rho''''(b_0)|<4$ bound follows from $.68<b_0<.69$, $3.14<\pi<22/7$, and the entirely rational comparison

$$\frac{225(22/7)(11/14)^4(69/100)^2}{[1+(157/200)(17/25)^2]^7}<16.$$

The left side bounds $|\rho''''(b_0)|^2$ above. Taylor's integral remainder is

$$\rho(k)-T_3(k;m)=\frac{(k-m)^4}{6}\int_0^1(1-r)^3\rho''''(m+r(k-m))\,dr.$$

The entire segment stays in $[A/h,B/h]$. Bound its derivative between $-M_+$ and $-M_-$, use $\int_0^1(1-r)^3\,dr=1/4$, and integrate against $W\ge0$. The signed fourth power works identically on both sides of the midpoint. All integrals are finite since $D<1$. □

For a single source panel of length $L = D/h - b_0 > 0$, let $w = \int_{hb_0}^{D} W$. Then $\mu_4 \le w(L/2)^4$ and

$$\widetilde{B} - \mu_4/6 \le B \le \widetilde{B}, \qquad \mu_4/6 \le \frac{wL^4}{96} \le .0075L^4 < .008L^4. \tag{4.2}$$

The source's error is $.008L^4$; the last strict comparison alone is minor. The useful change is retaining the actual fourth moment and branch mass rather than replacing them by endpoint distance and total mass .72. For a degenerate panel $L = 0$, set the error to zero without evaluating derivatives at an artificial midpoint.

**Corollary 4.2** (Certified subdivision)**.** *Split the active $k$ interval into $n$ equal panels of length $L/n$, retaining the exact $\rho$ branch boundaries. The sum of their cubic midpoint approximations has one-sided error at most*

$$E_n = \frac{1}{24}\sum_i M_{+,i}\mu_{4,i} \le \frac{wL^4}{96n^4}.$$

*Proof.* Apply Theorem 4.1 panelwise; use $M_{+,i} < 4$, $\mu_{4,i} \le w_i(L/(2n))^4$, and $\sum_i w_i = w$. This bounds the refined error by a fourth-order envelope; it does not assert that the actual errors at two mesh sizes have exactly an $n^{-4}$ ratio. □

### 4.3 Exact moments and cancellation control

Define $L_n(z) = \int_0^z X^n/\sqrt{1-X^2}\,dX$ and $J_j(z) = \int_0^z X^j W(X)\,dX$. With $\beta_z = \sqrt{1-z^2}$,

$$L_0(z) = \arcsin z, \quad L_1(z) = 1 - \beta_z,$$
$$L_n(z) = \{(n-1)L_{n-2}(z) - z^{n-1}\beta_z\}/n, \quad n \ge 2, \tag{4.3}$$
$$J_j(z) = \lambda L_{j+1}(z) - (1-\lambda)\{z^j/\beta_z - \mathbf{1}_{j=0} - jL_{j-1}(z)\}. \tag{4.4}$$

The $jL_{j-1}$ term is absent for $j = 0$. Integration by parts gives (4.3); differentiating $X^j/\beta_X$ gives (4.4), including the boundary term at zero. The additional fourth moment uses only $J_4$ and $L_5$, one degree beyond the source's cubic evaluator:

$$\mu_j = \sum_{r=0}^{j}\binom{j}{r}(-m)^{j-r}h^{-r}[J_r(B) - J_r(A)]. \tag{4.5}$$

Odd moments need not vanish because $W$ is not symmetric about $m$. Removing them solely because the panel is symmetric would be incorrect.

Formula (4.5) may cancel substantially, especially after subdivision. Outward intervals must be retained throughout. Since $\mu_4 \ge 0$ analytically, intersecting its enclosure with $[0, \infty)$ is legitimate; replacing an uncertain upper endpoint by zero is not. A second valid upper bound is $w(B/h - A/h)^4/16$. Taking the smaller of the two certified upper bounds improves reliability. Moment conditioning and extra arcsine evaluations are costs of the refinement; no timing advantage is claimed.

## 5 Curvature bounds that survive switches

### 5.1 The baseline has a logarithmic concavity correction

For fixed $X$, every selected smooth branch of (3.5) has second $h$ derivative at most $2/h^2$. For the quadratic branch this is $6\pi(X/h)^2/h^2 \le 6/(\pi h^2) < 2/h^2$; for the linear branch it is $2(X/h)/h^2 < 2/h^2$.

Concavity of $\rho$ gives

$$\frac{d^2}{dh^2}\rho(X/h) = \{k^2\rho''(k) + 2k\rho'(k)\}/h^2 \le 2/h^2,$$

since $k\rho'(k) \le \rho(k) \le 1$. The same holds for $t$. As $h$ increases, the branches occur in the order $\rho$, linear, quadratic. At $k = b_0$, $\rho'(b_0) = 2/(\pi+1) < 1$, and at $k = a_0$ the derivative changes from $-a_0/h$ to $-2a_0/h$. Both jumps decrease the first derivative. There is no positive distributional atom.

**Proposition 5.1.** *$B(h) + 2\log h$ is concave on $[.64, 2.25]$.*

*Proof.* Adding $2\log h$ cancels the upper smooth bound $2/h^2$ and leaves the nonpositive derivative jumps unchanged for each selected branch and for $t$. The weights $\ell + \int_0^D W = 1$ are nonnegative. Thus their weighted sum plus the affine term in (3.5) is concave. □

This is a direct sharpening of the source's constant correction $B - h^2/a^2$ on $[a, b]$; it does not assume that a nonsmooth minimum can be twice differentiated across a tie.

## 5.2 A smaller constant for the gain

**Proposition 5.2.** *The gain $S$ has upper second derivative bound*

$$\gamma = .26866728 < .269 \tag{5.1}$$

*on $[1.02, 1.62]$: $S - \gamma h^2/2$ is concave.*

*Proof.* Fix $y \ge 0$, $w = u - 1 + R - Qy$, $R = 2ht/\pi$, and $F_y = \ell y^2 + C_0 \min(d_m, w_+)^3$. Differentiating the equation for $t$ gives

$$R' = \frac{2t^2}{\pi(2-t)}, \quad R'' \le 0, \quad \frac{Q'}{Q} = \frac{2h}{\pi + 4h^2}, \quad \frac{Q''}{Q} = \frac{2\pi - 4h^2}{(\pi + 4h^2)^2}.$$

The source's rational estimates are $.64 < t < .8$, $.19 < R' < .34$, $R < .671$. On an active piece $0 < w < d_m$, $0 \le Qy < P < .36$, $Q'/Q \le 1/(2h)$, $Q''/Q \ge -1/(4h^2)$. On the constant branch of $u$, $0 < w' < .34$; on its reciprocal branch, whose switch is above 1.4, $-.50 < w' < .34$. Both have $w'' < .8$. Also $0 < C_0 \le .36$, $0 \le C_0' \le 1.08$, $0 \le C_0'' \le 2.16$ on smooth pieces. Hence

$$\begin{aligned}(C_0 w^3)'' &= C_0'' w^3 + 6C_0' w^2 w' + 6C_0 w (w')^2 + 3C_0 w^2 w'' \\ &\le 2.16 d_m^3 + 6(1.08) d_m^2 (.34) + 6(.36) d_m (.50)^2 + 3(.36) d_m^2 (.8) \\ &= .26866728.\end{aligned}$$

On the zero branch the derivative bound is zero; on the capped branch it is at most $2.16 d_m^3 < \gamma$.

At switches in $u$ or $C_0$, their derivative jumps are nonpositive. On $0 < w < d_m$ the resulting jump is $w^3 \Delta C_0' + 3C_0 w^2 \Delta w' \le 0$. At $w = 0$ both the cubic and its first derivative vanish. At $w = d_m$, the incoming and outgoing derivatives of the capped cubic are $3d_m^2 \max(w_-', 0)$ and $3d_m^2 \min(w_+', 0)$; their difference is nonpositive, even at simultaneous switches. Algebraicity gives finitely many threshold points on each fixed-$y$ branch unless equality persists identically, in which case the derivatives match as stated. Thus $F_y - \gamma h^2/2$ is concave. The pointwise infimum of concave functions is concave: its hypograph is their intersection. Since $0 \le S \le C_0 d_m^3$, the infimum is finite, proving the assertion. □

The source computes the same numerical sum but then rounds its curvature constant up to .4. Retaining it is a modest realized tightening, not a new geometric estimate. For convenient rational computation below we use $\bar\gamma = .269 > \gamma$.

# 6 Continuous certificates from finite knot enclosures

## 6.1 A logarithmic lower envelope

Let $f = B$ on an exterior interval, or $f = B + S$ on an interior interval. Set $\chi = 0$ or 1 accordingly, and suppose certified lower bounds $L_a \le f(a)$, $L_b \le f(b)$ are available. Proposition 5.1 and Proposition 5.2 give, for every $h \in [a, b]$,

$$f(h) \ge (1-s)L_a + sL_b - 2\{\log h - (1-s)\log a - s\log b\} - \frac{\chi\gamma}{2}(h-a)(b-h), \quad s = \frac{h-a}{b-a}. \tag{6.1}$$

This is rigorous but involves logarithms. The maximum logarithmic loss occurs at the logarithmic mean $h_* = (b-a)/\log(b/a)$, since the derivative of the bracket is $1/h - \log(b/a)/(b-a)$. Directed enclosures for logarithms would be required for a machine certificate based on this exact envelope. We do not call ordinary floating-point logarithms certified.

## 6.2 A rational endpoint-sensitive alternative

The rational upper curvature bound

$$K = 2/a^2 + \chi\bar{\gamma}, \qquad \bar{\gamma} = .269, \tag{6.2}$$

already improves the source's interior constant 2.4. Set $\Delta = b - a$, $C = K\Delta^2/2$, and $d = L_b - L_a$. Concavity yields

$$f(a + s\Delta) \ge q(s) := L_a + (d - C)s + Cs^2, \qquad 0 \le s \le 1. \tag{6.3}$$

The exact rational minimum is

$$Q_{a,b} = \begin{cases} L_a, & d \ge C, \\ L_b, & d \le -C, \\ L_a - (d-C)^2/(4C), & -C < d < C. \end{cases} \tag{6.4}$$

For $C = 0$, use $\min(L_a, L_b)$; here all nondegenerate intervals have $C > 0$. The source's bound $\min(L_a, L_b) - C/4$ discards the slope information. Equation (6.4) follows simply by checking the stationary point $(C - d)/(2C)$ and both endpoints, so it never gives a worse bound for the same $K, L_a, L_b$.

## 6.3 Bernstein form without pretending the integrand is polynomial

The private power-to-Bernstein identity [6] gives the degree-two coefficients

$$\beta_0 = L_a, \qquad \beta_1 = (L_a + L_b - C)/2, \qquad \beta_2 = L_b.$$

For $s \in [0, 1]$, $q(s) = \sum_{j=0}^{2} \beta_j \binom{2}{j} s^j (1-s)^{2-j}$ lies above $\min \beta_j$. These coefficients may be subdivided exactly by de Casteljau averaging. A negative coefficient is inconclusive, not a counterexample; the exact quadratic minimum is preferable at degree two. Crucially it is $q$, a proved lower envelope, which is polynomial. Neither $\rho$, $B$, nor the minimizing branch of $S$ has been declared polynomial.

### 6.4 Error ledger and conditional geometric conclusion

At a knot, a safe baseline lower endpoint is

$$L_h^B = \underline{\widetilde{B}}(h) - \overline{E}_h.$$

Here the first quantity already includes every outward arithmetic and arcsine-tail error, while $E_h$ is the analytic truncation error, either the source's $.008L^4$ or Theorem 4.1. Then $L_h = L_h^B + \chi\underline{S}(h)$. Do not deduct the same interval rounding width twice, and do not omit the analytic error merely because the algebraic approximation was evaluated exactly.

**Corollary 6.1** (Precisely scoped exclusion)**.** *If all* 23 *intervals in the source knot partition have positive certified lower bounds for the appropriate $f$, and if the imported geometric statements through the nesting area bound hold for every nonstandard minimizer, then no such minimizer exists at $V_2$. Combined with the imported first-endpoint and four-vertex exclusions and two-volume reduction, this implies* (2.1) *and its equality classification.*

*Proof.* Every surviving candidate has $h \in [.64, 2.25]$ and $A \le 1$. The imported bound gives $A - 1 \ge f(h) > 0$, a contradiction. The remaining claims follow through the dependency chain in Section 2, not through scalar arithmetic alone. □

## 7 Independent arithmetic: protocol and results

### 7.1 What is and is not being executed

The local snapshot contains the paper, seven section files, two figure files, references, and README; no source-supplied executable verifier is present in that manuscript directory. Its sentence that reproducible exact calculations are included does not justify claiming that an unavailable executable was run. The independent evaluator supplied with this report implements the explicit arithmetic prescription and separately checks our refinements. It is not a formal proof assistant.

The author executed `verify_scalar.py` with the bundled Python runtime. It passed rational Machin verification of the source's $\pi$ interval; all 24 outward baseline knot enclosures; all 16 gain values; the 23 original interpolation bounds; signed moment enclosures with one, two, and four panels; and rational endpoint-sensitive gaps. It also checked positivity of the quadratic lower envelopes after four binary Bernstein subdivisions (16 leaves) on every interval. All acceptance comparisons use integers or exact rational fractions. Decimal displays in JSON are descriptive only.

The full scalar record is `_audit/M682-author-execution.json`; the exact arithmetic kernel and formulas are in the report directory. This is an independently authored implementation of the source's displayed prescription, not execution of a source-supplied verifier. A short self-test suite checks the interval primitives against rational examples, including negative products, reciprocals, even powers straddling zero, and integer square-root inequalities. These tests do not establish formal correctness of the Python implementation.

### 7.2 Arithmetic design

Intervals have integer endpoints at scale $N = 10^{40}$. Addition and negation are exact. Multiplication minimizes and maximizes the four endpoint products and rounds outward; reciprocal requires exclusion of zero; square root uses integer-square comparisons. Even powers crossing zero have lower endpoint zero. Minima act coordinatewise, so dependence affects width, not validity.

For $0 \le z \le D$, put $m(z) = z/\sqrt{2+2\sqrt{1-z^2}}$, $j = m(m(z))$. Then $\arcsin z = 4\arcsin j$, $j < .36$, and for truncation index $n$,

$$4\sum_{r=0}^{n} \frac{\binom{2r}{r} j^{2r+1}}{4^r(2r+1)} \le \arcsin z \le 4\sum_{r=0}^{n} \frac{\binom{2r}{r} j^{2r+1}}{4^r(2r+1)} + \frac{4j^{2n+3}}{1-j^2}.$$

The replay uses the source's $n = 10$; tighter moment experiments may use $n = 20$, reported separately. The source's $\pi$ enclosure is checked against alternating rational arctangent sums in Machin's identity. No transcendental library call is needed to certify these intervals.

For the gain use

$$q_* = \frac{2P}{1+\sqrt{1+6C_0Q^2P/\ell}}, \qquad S = \min\{C_0 d_m^3, \ell((P-q_*)/Q)^2 + C_0 q_*^3\}. \tag{7.1}$$

Here $P, Q, C_0 > 0$ on the interior range. To check caps, put $w = P - Qy$. The uncapped objective $\ell(P-w)^2/Q^2 + C_0 w^3$ is strictly convex on $[0, P]$, with critical point $q_*$. The zero branch has value at least its value at $w = 0$. If $P \ge d_m$, the capped branch minimizes at $w = P$ with value $C_0 d_m^3$. When $q_* > d_m$, even its infeasible cubic value exceeds the cap. When $P < d_m$, the cap value exceeds the feasible value at $P$, so cannot corrupt the minimum. Formula (7.1) therefore includes all cap and tie cases.

### 7.3 All interval margins, not selected samples

Table 1 gives integer lower bounds in units of $10^{-6}$. The first numerical column is the printed source claim, reproduced by our run. The second uses exactly the source's rounded $b_h, s_h$ and original Taylor-error subtraction, changing only the curvature constant and the minimization of the quadratic envelope. The third uses the four-panel moment lower endpoints and directly enclosed gain values as well. It therefore combines tighter quadrature with recovery of the source table's rounding slack; it must not be attributed to the moment error alone.

Table 1: Certified uniform lower bounds on all source intervals.

| Interval | Source | Same knots, refined | Four-panel, refined |
|---|---|---|---|
| [.64, .74] | 4416 | 5578 | 8120 |
| [.74, .84] | 2298 | 3943 | 4752 |
| [.84, .92] | 1859 | 3021 | 3302 |
| [.92, .98] | 1313 | 2008 | 2136 |
| [.98, 1.02] | 967 | 1316 | 1391 |
| [1.02, 1.06] | 1037 | 1336 | 1399 |
| [1.06, 1.10] | 581 | 848 | 894 |
| [1.10, 1.14] | 280 | 511 | 553 |
| [1.14, 1.18] | 145 | 327 | 364 |
| [1.18, 1.22] | 137 | 280 | 311 |
| [1.22, 1.26] | 137 | 329 | 359 |
| [1.26, 1.30] | 209 | 412 | 441 |
| [1.30, 1.34] | 269 | 511 | 543 |
| [1.34, 1.38] | 379 | 644 | 676 |
| [1.38, 1.42] | 369 | 651 | 677 |

| Interval | Source | Same knots, refined | Four-panel, refined |
|---|---|---|---|
| $[1.42, 1.46]$ | 90 | 438 | 468 |
| $[1.46, 1.50]$ | 90 | 333 | 362 |
| $[1.50, 1.54]$ | 100 | 510 | 537 |
| $[1.54, 1.58]$ | 520 | 1000 | 1030 |
| $[1.58, 1.62]$ | 1470 | 1950 | 1976 |
| $[1.62, 1.70]$ | 250 | 860 | 878 |
| $[1.70, 1.90]$ | 1899 | 5360 | 5379 |
| $[1.90, 2.25]$ | 14266 | 22750 | 22763 |

The new worst interval is $[1.18, 1.22]$. Before display rounding, its bounds are greater than .0002803536 and .0003114899 in the second and third columns. These are lower bounds, not computed exact minima of $B + S$. With the unchanged 24 knots, the same-knot uniform bound improves by more than a factor of three. No fewer-knot or runtime-speedup claim follows.

The parent independently reconstructed the source scalar prescription in a separate program, `_audit/M682-parent-finite-20261008/check_scalar.py`. Its `scalar-results.json` records 238 exact/outward checks passing, including all 24 baseline knots, 16 gain knots, 23 source gaps, and both complete source slack ranges. Its same-rounded-knot margin is greater than .00028042024020. The small difference from our .00028035369815 is fully explained by the curvature constant: that reconstruction uses $\gamma = .26866728$, whereas Table 1 uses the slightly larger rational convenience bound $\bar{\gamma} = .269$. The parent also replayed our unchanged verifier, reproducing both our same-knot and four-panel margins. That replay is not a third independent implementation. Agreement of two separately authored programs implementing the same published formulas reduces implementation risk but does not independently validate every shared analytic premise or constitute formal verification.

The complete source slack statements also pass:

$$1.10 < 10^5 \widetilde{B}(h) - b_h < 1.95, \qquad 1.00 < 10^5 S(h) - s_h < 1.66.$$

The extrema of the outward enclosures, displayed approximately, are $[1.10434669, 1.94465984]$ and $[1.00330385, 1.65026165]$, respectively. At $h = 1.46$, the source Taylor interval is absent and the capped value is exactly $.36(.21)^3 = .00333396$. The negative baseline there is not a rounding artifact; nesting is necessary for this argument.

Table 2 isolates the analytic truncation bounds, rounded upward. The refined columns use the conservative $M_+ < 4$ bound, not a numerically fitted derivative constant.

Table 2: Upper bounds for baseline Taylor error.

| $h$ | Source error | One-panel moment | Four-panel moment |
|---|---|---|---|
| .64 | .004474250 | .000843270 | $2.719464\,10^{-6}$ |
| 1.02 | $5.553350\,10^{-5}$ | $6.642433\,10^{-6}$ | $2.401153\,10^{-8}$ |
| 1.42 | $4.397877\,10^{-10}$ | $4.226263\,10^{-12}$ | $5.837430\,10^{-14}$ |
| 1.46 | 0 | 0 | 0 |

### 7.4 Independent symbolic and diagnostic checks

A separate script `check_identities.py` ran with Python 3.10 and installed SymPy 1.11.1. It verified 17 identities: the third and fourth derivatives of $\rho$, the weight derivative, the derivative and

zero endpoint of each $J_0, \dots, J_4$, the gain-curvature arithmetic, the defining equation for $t$, $R'$, and $Q''/Q$. The record is `_audit/M682-author-symbolic.json`.

Seven branch-split 70-decimal-digit numerical quadratures were also run at $h = .64$, 1.02, 1.18, 1.42, 1.46, 1.62, and 2.25. These are explicitly nonrigorous diagnostics and do not enter any certified comparison. For example they give $B(.64) \approx .01712551408346$ and a one-panel actual error approximately .00032758936173, consistent with the one-sided certified interval. Such agreement tests implementation plausibility but does not replace outward arithmetic.

### 7.5 Independent geometric review and platform record

The separate geometric audit in `_audit/M682-independent-20261008/audit-notes.md` reports no unresolved P1/P2 finding in its bounded reading of reduction, exclusions, and equality cases. It separately checks the distinction between original-region reflection and a symmetric replacement, the three-vertex side choice, singular slice levels, both strip orientations, degenerate caps, and transition co-minimizers. It explicitly does not execute the scalar certificate. We retain that limitation, rather than converting the audit's absence of findings into formal global verification.

That sidecar records direct public-pin SHA-256 agreement for all 11 TeX inputs. Our local snapshot is not a Git checkout; its failed `git rev-parse HEAD` is not a provenance verification. The manifest binds the actual read inputs, the geometric sidecar, and the separate arithmetic records. The parent's full-draft and code review found no issue with the local signed-moment, curvature, logarithmic-envelope, or quadratic/Bernstein derivations; this remains a bounded mathematical review.

A separate focused follow-up, `_audit/M682-independent-20261008/focused-followup.md`, found no unresolved P1/P2 issue in the signed fourth-moment theorem, panel corollary, or sharper switch-safe gain bound. Its `check-followup.py` parsed the rounded source table and passed 82 exact rational assertions, matching every parent chord fraction and explicitly verifying the .00028 threshold for both curvature constants. It did not rerun integrals, evaluate panel quadrature, or inspect our interval implementation; the four-panel result remains supported by our saved execution and the parent's replay of that inspected code. These distinct scopes are preserved in the evidence records.

The native LaTeX compiler was attempted on the saved source and returned `compile-failed`, with log "Unable to find standard directories for platform." The editor-opening request returned `queued`, not a confirmation that an editor was visible or that compilation succeeded. The installed MiKTeX helper is used solely for the separately requested on-disk PDF and visual QA; its actual outcome is recorded with the final export.

## 8 Benefits, limitations, and research directions

The signed remainder provides a two-sided enclosure and a better lower bound using the same cubic derivatives and one extra moment degree. Subdivision has an explicit error envelope rather than an empirical convergence claim. The branch jump analysis permits continuous certification without a hidden $C^2$ assumption. Endpoint-sensitive interpolation uses information already available in the knot table. These benefits are local and verifiable.

The costs are cancellation in centered moments, additional endpoint evaluations, and a larger implementation surface. The source's proof already has a strictly positive claimed scalar margin; a larger certified margin does not enlarge the geometric theorem's volume range, classify new geometries, or replace the nesting argument. No speedup is measured, no proof of all code correctness is supplied, and global geometric correctness is not established by these tests.

Useful further work is to exploit the logarithmic envelope with certified logarithms; optimize panel sizes using rigorous $M_{+,i}\mu_{4,i}$ bounds; use a weight-adapted center while retaining all odd moments; or formalize the small arithmetic kernel and branch logic. Anisotropic rectangular tori would require new geometric constants and a new nesting analysis, not substitution of a different $\pi$ or period. Optimizing $\lambda$ is a conditional design problem: the location of the flat minimum, weight positivity, geometric derivative estimates, and every knot certificate would need recomputation. No conjectural improvement is counted in the contribution assessment.

## 9 Public literature check and assessment

A public-title search on October 7, 2026 and direct primary-source access confirmed Milman's article [2], published online November 27, 2025 and in volume 79 (2026), pages 1012–1072. Its stated cubic spherical range reaches 0.120582, and it discusses reductions of the conjecture and the obstruction from non-nested base minimizers. This precedes and is distinct from the pinned OpenAI full-classification claim. Public searches also found Milman's August 2026 *Spectral and Isoperimetric Bounds on Flat Tori* [3], a different parameter-comparison problem; it is not evidence validating the present full profile. Direct web opening of the pinned GitHub directory and of the HPRR DOI returned tool errors; local source reading still succeeded. Search coverage is not an exhaustive novelty review.

The internal policy rates realized contribution relative to the pinned manuscript, not report length or applicability. C2 is appropriate for the proved and executed local refinements: the scalar margin is enlarged and the analytic error is reduced, but the source already has a successful scalar strategy and no new geometric conclusion is obtained. We retain its geometric claims as dependencies. The assessment record binds this judgment to the final source hash and executed evidence. No credit is assigned for imported profile theorems or unexecuted future work.

## A Read locators and reproducibility contract

The final nested manifest records full input paths, SHA-256 hashes, and read locators. Source paths below are relative to the manuscript's `build` directory; all were read in full.

| Source | Substantive content retained |
|---|---|
| `paper.tex` | Introduction, main theorem, completion, normalization and equality claims. |
| `sections/preliminaries.tex` | Classical structure, competitors, profile supports, two-volume reduction. |
| `sections/geometry.tex` | Reflection, nodal domains, monotonicity, two vertex patterns. |
| `sections/sections.tex` | Finite exceptional levels, absolute continuity, flux, projection, coarea. |
| `sections/early-exclusions.tex` | Four-vertex exclusion; first endpoint; second-endpoint parameter domain. |
| `sections/nesting.tex` | Jensen flux, bottom calibration, both strip orientations, all slice types, moments, gain. |
| `sections/elementary-estimates.tex` | Both scalar ranges and their rational comparisons. |

| Source | Substantive content retained |
|---|---|
| `sections/numerics.tex` | Entire layer cake, curvature proofs, Taylor error, moment recurrence, cap evaluation, arithmetic, both tables. |
| `references.tex`; both figure sources | Public references and the precise geometric meanings of the schematic drawings. |

Private reads were actual text, not catalogue-map substitutes: volume 1 revised chapter 3, lines 1–245, especially `thm:ram-local`, `eq:ram-integral-remainder`, `eq:ram-remainder-bound`, and `thm:degree-closure`; volume 16 clearance chapter, lines 1–156, especially the conversion proof, convex-hull bound and subdivision limitations; TR-item-047-astra, lines 131–225 and 341–412, especially the replayable gap and inexact Gram/identity certificate. Bibliographic identifiers are local confidential references rather than public links.

Reproducibility artifacts include the read-locator manifest, independently authored scalar and symbolic scripts, exact interval/fraction outputs, separate parent arithmetic and review records, native-platform diagnostics, and disk-export visual QA. The manifest stores nested path/SHA-256 pairs for the inputs and final TeX/PDF. The finite results are reproducible from the saved local inputs without sending confidential material to external services.

## References


[1] OpenAI, *The Isoperimetric Conjecture for the Cubic Flat Three-Torus*, September 24, 2026. Pinned commit stated in Section 1. Public pinned manuscript.

[2] E. Milman, *Isoperimetric inequalities on slabs with applications to cubes and Gaussian slabs*, Comm. Pure Appl. Math. **79** (2026), 1012–1072. doi:10.1002/cpa.70020; arXiv:2403.06602.

[3] E. Milman, *Spectral and Isoperimetric Bounds on Flat Tori*, August 13, 2026. arXiv:2608.13052. Public abstract checked for scope only.

[4] L. Hauswirth, J. Pérez, P. Romon, A. Ros, *The periodic isoperimetric problem*, Trans. Amer. Math. Soc. **356** (2004), 2025–2047. doi:10.1090/S0002-9947-03-03362-2. Used as an imported source dependency, not independently re-audited here.

[5] M. Subbarao, private Ram Calculus corpus, volume 1 revision 2, *Local Moments and Ram Differential Operators*, chapter 3, local representation and polynomial-closure passages. Confidential local source.

[6] M. Subbarao, private Ram Calculus corpus, volume 16, clearance chapter, *Polynomial enclosures in Bernstein form* and subdivision passages. Confidential local source.

[7] M. Subbarao, private report `TR-item-047-astra.tex`, *Global Optimality Sandwich* and *Residual-Aware SOS Lower Bounds*. Confidential local source.